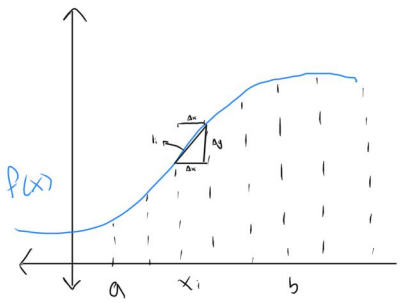
Lecture 5

Arc length



$$\Delta x = \frac{b-a}{n}$$

l_i : length of tangent at x_i

* $l_i^2 = (\Delta x)^2 + (\Delta y)^2$
* $f'(x_i) = \frac{\Delta y}{\Delta x} \Rightarrow \Delta y = f'(x_i)\cdot\Delta x$

$$l_i^2 = (\Delta x)^2 + [f'(x_i)\cdot\Delta x]^2$$
$$= [1 + f'(x_i)^2]\cdot(\Delta x)^2$$
$$l_i = \sqrt{1+f'(x_i)^2}\cdot\Delta x$$

approx length $= \sum_{i=1}^{n} l_i$

exact length $= \lim_{\substack{n\to\infty \\ \Delta x\to 0}} \sum_{i=1}^{\infty} l_i$

$$l = \int_{x=a}^{x=b} \sqrt{1+[f'(x)]^2}\,dx$$

ex: calculate the arc length of $y = 4\cdot x^{3/2}$

$x=1$ to $x=4$

$$l = \int_a^b \sqrt{1+[f'(x)]^2}\,dx$$

$f'(x) = 6\cdot x^{1/2}$

$$l = \int_{x=1}^{x=4} \sqrt{1+36x}\,dx \Rightarrow u = 1+36x$$

$\frac{du}{dx} = 36$

$dx = \frac{du}{36}$

$x=1 \to u=37$

$x=4 \to u=145$

$$l = \int_{37}^{145} \sqrt{u}\,\frac{du}{36}$$
$$= \frac{1}{36}\int_{37}^{145} u^{1/2}\,du$$
$$= \frac{1}{36}\left[\frac{2}{3}u^{3/2}\right]_{37}^{145}$$
$$= \frac{1}{36}\cdot\left[\frac{2}{3}(145)^{3/2} - \frac{2}{3}(37)^{3/2}\right]$$
$$\simeq 28.17$$

Hydrostatic pressure

$$\text{Pressure} = \frac{\text{force}}{\text{area}}$$

→ weight of fluid (w) on this plate

$= \rho V g$

→ $V = A d$

$w = \rho A d g$

→ pressure (p) on this plate

$$= \frac{W}{A} = \frac{\rho A g d}{A} = \rho g d \quad (N/m^2 = \text{pascal})\ (lb/ft^2)$$

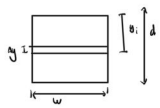
Hydrostatic pressure at depth $d = \rho g d$ from the top of the water

What is the hydrostatic force on the sides

→ hydrostatic pressure at depth $y_i = \rho g y_i$

→ Area of the strip $w\Delta y$

→ hydrostatic force on this strip = pressure · area $= \rho g y_i w \Delta y$



→ total (approx) hydrostatic force on this side

$$= \sum_{i=1}^{n} \rho g y_i w \Delta y$$

→ total (exact) hydrostatic force on this side

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \rho g y_i w \Delta y$$
$$= \int_0^d \rho g w y\,dy$$

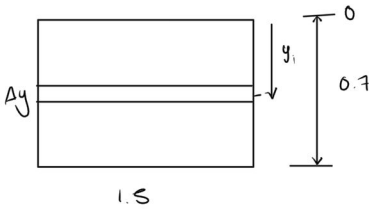
ex: A fish tank is 1.5m long and 1m wide by 0.7 m deep

a) find hydrostatic force on the bottom

b) find hydrostatic force on each of 4 sides

a) F = weight of the water

$$= \rho g d A$$
$$= 1000 \cdot 9.8 \cdot 0.7 \cdot 1 \cdot 1.5$$
$$= 10\,290\ N$$



b) → choose one side

Force on this strip

$= \rho g d A$

$= 1000 \cdot 9.8 \cdot y_i \cdot 1.5 \cdot \Delta y$

total force on that side

$$= \int_0^{0.7} 14700\,y\,dy = 3601.5\,N$$

→ same force on the opposite side

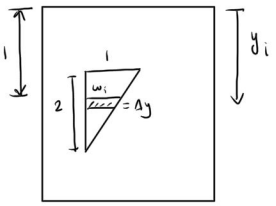
→ the other 2 sides $\int_0^{0.7} 1000\cdot 9.8\cdot 1\cdot y\,dy$

ex: - suppose a vertical plate is submerged 1m under the water, where the plate has dimensions as in figure. What is the hydrostatic force on the plate?

the force on the strip

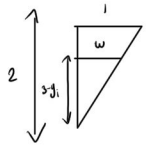
$F_i = \rho g d A$

$= \rho g y_i w_i \Delta y$



by similarity

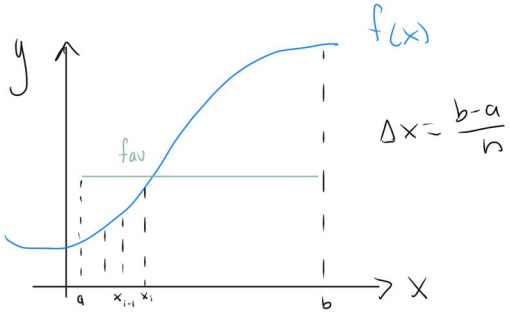
$$\frac{w_i}{1} = \frac{3-y_i}{2}$$
$$w_i = \frac{3}{2} - \frac{y_i}{2}$$



$$F_i = \rho g y_i \left(\tfrac{3}{2} - \tfrac{y_i}{2}\right)\Delta y$$
$$F = \int_1^3 9800\,y\left(\tfrac{3}{2} - \tfrac{y}{2}\right)dy$$
$$= \int_1^3 9800\left(\tfrac{3}{2}y - \tfrac{y^2}{2}\right)$$
$$9800\left[\tfrac{3}{4}y^2 - \frac{y^3}{6}\right]_1^3$$

Lecture 6

Average Value



$\Delta x = \frac{b-a}{n}$

Split interval $[a,b]$ into n subintervals

Choose $x_i^* \in [x_{i-1}, x_i]$ $\quad [f(x_1^*), f(x_2^*), f(x_3^*), \dots, f(x_n^*)]$

$$\text{approx } f_{av} = \frac{\sum_{i=1}^{n} f(x_i^*)}{n} = \frac{\sum_{i=1}^{n} f(x_i^*)}{\left[\frac{b-a}{\Delta x}\right]}$$

$$= \frac{1}{b-a} \cdot \sum_{i=1}^{n} f(x_i^*) \cdot \Delta x$$

$$\text{exact } f_{av} = \frac{1}{b-a} \lim_{\substack{n\to\infty \\ \Delta x \to 0}} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

$$= \frac{1}{b-a} \int_a^b f(x)\,dx$$

ex: find the average value of $f(x) = \sin(x)$ on $[0, \pi]$

$$f_{av} = \frac{1}{\pi - 0} \int_0^{\pi} \sin(x)\,dx$$

$$\frac{1}{\pi} * -\cos(x)\Big|_0^{\pi}$$

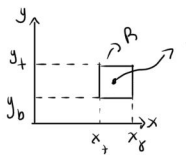
$$\frac{1}{\pi} * [-\cos(\pi) + \cos(0)]$$

$$= \frac{2}{\pi}$$

Moment & center of mass

Moment: the tendency to tip over or rotate around an axis.

* Assume we have a rectangular plate of uniform density ρ is situated in the x-y plane



center of mass $(\bar{x}, \bar{y}) = \left(\frac{x_t + x_g}{2}, \frac{y_b + y_t}{2}\right)$

The mass of R $= \rho * \text{area}$

The moment of R about x-axis

M_x = mass * (signed) distance from center of mass to x-axis

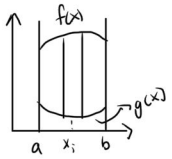
$= \rho * A * \bar{y}$

The moment of R about y-axis

$M_y = \rho * A * \bar{x}$

* if the area is not regular [no formula]

The moment of the region R bounded by $f(x), g(x), x=a, x=b$ about x-axis & y-axis



→ split the region into n rectangles each of width Δx

→ find the center of mass $\left(x_i, \frac{f(x_i) + g(x_i)}{2}\right)$

→ find moment of the rectangle about x-axis

$$(M_x)_i = \rho * A_i * \text{distance}_x$$

$$= \rho * (f(x_i) - g(x_i))\Delta x * \frac{f(x_i) + g(x_i)}{2}$$

→ find approximate moment

$$\text{approx } M_x = \sum_{i=1}^{n} (M_x)_i$$

→ find exact moment

$$M_x = \lim_{n\to\infty} \sum_{i=1}^{n} (M_x)_i$$

$$= \int_a^b \rho \cdot [f(x) - g(x)] * \frac{[f(x) + g(x)]}{2}\,dx$$

$$\frac{1}{2} \int_a^b \rho [f^2(x) - g^2(x)]\,dx$$

→ find moment of the rectangle about y-axis

$$(M_y)_i = \rho * A_i * \text{distance}_y$$

$$= \rho * [f(x_i) - g(x_i)] * \Delta x * x_i$$

$$\text{approx } M_y = \sum_{i=1}^{n} \rho\, f(x_i) - g(x_i) - \Delta x * x_i$$

$$\boxed{M_y = \int_a^b \rho \cdot x \cdot [f(x) - g(x)]\,dx}$$

→ what is the center of mass of R

we know that:

M_x = mass * (signed) distance from center of mass to x-axis

$= \text{mass} * \bar{y}$

$M_y = \text{mass} * \bar{x}$

if we have M_x and M_y we can get center of mass

$\text{mass} = \rho * A$

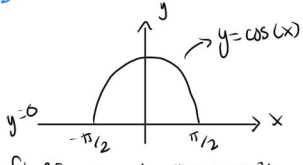
$= \int_a^b \rho [f(x) - g(x)]\,dx$

$$\boxed{\bar{x} = \frac{M_y}{\text{mass}}, \quad \bar{y} = \frac{M_x}{\text{mass}}}$$

* Note: if ρ is not fixed [depend on x and/or y] it has to be inside the integral

→ if fixed it can be extracted from integral

ex: find the moment M_x & M_y & center of mass (centroid) of a flat plate of uniform density $\rho = 1$, shaped like the region R bounded by $y = \cos(x)$ and the x-axis for $-\frac{\pi}{2} \le x \le \pi/2$



$$M_x = \int_{-\pi/2}^{\pi/2} \frac{1}{2} \rho [\cos^2(x) - 0]\,dx$$

$$= \frac{1}{2} \int_{-\pi/2}^{\pi/2} \cos^2(x)\,dx$$

$$= \frac{1}{2} \int_{-\pi/2}^{\pi/2} \frac{1}{2} + \frac{1}{2}\cos(2x)\,dx$$

$$= \frac{1}{4} \left[x + \frac{1}{2}\sin(2x)\right]_{-\pi/2}^{\pi/2}$$

$$= \pi/4$$

$$M_y = \int_{-\pi/2}^{\pi/2} \rho * x [\cos(x) - 0]\,dx$$

$$= \int_{-\pi/2}^{\pi/2} (x \cos(x))\,dx$$

$$= x\sin x\Big|_{-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2} \sin(x)$$

$$= [x\sin x + \cos x]_{-\pi/2}^{\pi/2}$$

$$= 0$$

[integration by parts: $u = x$, $du = dx$; $dv = \cos x$, $v = \sin x$]

Centroid

$\text{Mass} = \rho \cdot A$

$$= \int_{-\pi/2}^{\pi} \cos(x) - 0\,dx$$

$$= \sin(x)\Big|_{-\pi/2}^{\pi/2} = 2$$

$\bar{x} = \frac{M_y}{\text{mass}} = 0$ & $\bar{y} = \frac{M_x}{\text{mass}} = \pi/8$

Midterm 1 exam Sheet

Lecture 1 - Improper intergrals

Definite intergrals

$\int_a^b f(x)\,dx$ → area between f & x-axis on $[a,b]$

if cts on $[a,b]$ → $\int_a^b f(x)\,dx = F(x)\Big|_a^b = F(b)-F(a)$

Improper integral of type 1

$a=-\infty$ and/or $b=\infty$

$\int_a^\infty f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$ $a,b\in\mathbb{R}$

$\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx$

$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^\infty f(x)\,dx$

* if cts on designated domain

general rule

$\int_1^\infty \frac{1}{x^p}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x^p}\,dx$

$= \lim_{t\to\infty} \frac{x^{-p+1}}{-p+1}\Big|_1^t$

$= \lim_{t\to\infty} \frac{t^{-p+1}-1}{-p+1}$

if $p>1$ the integral converge

if $p\le 1$ the integral diverge

$\lim_{t\to\infty} t^p = \begin{cases} 0 & p<0 \\ \infty & p>0 \end{cases}$

* with $\int_{-\infty}^\infty f(x)\,dx$ it only converges when both of $\int_{-\infty}^a f(x)\,dx$ & $\int_a^\infty f(x)\,dx$ converge.

Improper integral of type 2

discontinuity in domain

$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right)+C$

* discontinuity at b

$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$

* discontinuity at a

$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$

* discontinuity at c, $a<c<b$

$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

Lecture 2 - area between curves

rule

assume f,g are continuous on $[a,\infty)$

if $\int_a^\infty g(x)$ converge

and $0\le f(x)\le g(x)\ \forall\ x\ge a$

$\int_a^\infty f(x) \le \int_a^\infty g(x) < \infty$

then $\int_a^\infty f(x)$ converge

if $\int_a^\infty f(x)\,dx$ diverge

and $0\le f(x)\le g(x)\ \forall\ x\ge a$

$\infty = \int_a^\infty f(x) \le \int_a^\infty g(x)$

then $\int_a^\infty g(x)$ diverge

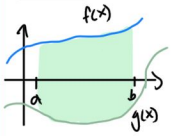
* useful integrals

$\int_1^\infty \frac{1}{x^p}\,dx$ → converges to $\frac{1}{p-1}$ if $p>1$; → diverges if $p\le 1$

$\int_0^1 \frac{1}{x^p}\,dx$ → converges to $\frac{1}{p-1}$ if $p<1$; → diverges if $p\ge 1$

$\int_0^\infty e^{-ax}\,dx$ → converges to $\frac{1}{a}$ if $a>0$; → diverges if $a<0$

areas

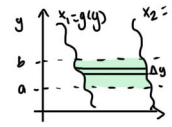


exact area

$= \int_a^b [f(x)-g(x)]\,dx$

$= \int_a^b |f(x)-g(x)|\,dx$

horizontal strips

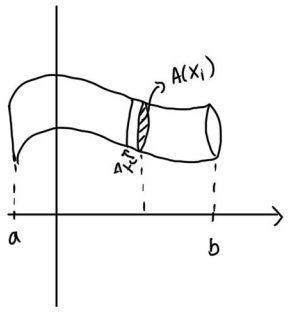


$\text{Area} = \int_a^b [f(y)-g(y)]\,dy$

horizontal strips (Δy) → this means write the functions as $x=f(y)$

everything in terms of y

Lecture 3 - Volumes ; cylindrical shells



Exact volume $= \int_a^b A(x)\,dx$

→ (volume by slices)

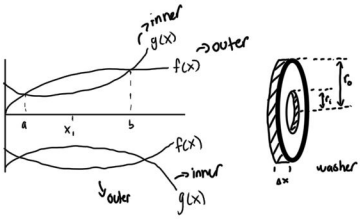
volume obtained by rotating region b/w 2 curves around an axis

$\text{Volume} = \int_a^b A(x)\,dx$

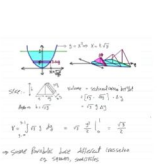
$= \int_a^b \pi\cdot[r_o^2(x)-r_i^2(x)]\,dx$

↳ area of washer

$= \int_a^b \pi[f^2(x)-g^2(x)]\,dx$

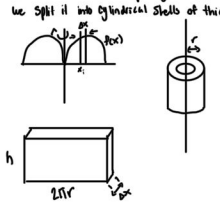


volume of a solid



volume by cylindrical shell

Idea: instead of decomposing the solid into slices we split it into cylindrical shells of thickness Δx (or Δy)



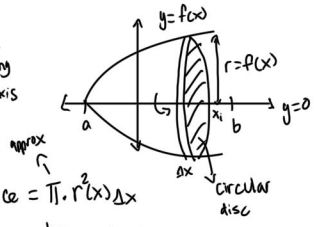
volume of the shell ↳ $2\pi r(x_i)\,h(x_i)\,\Delta x$

volume of all shells ↳ $\sum_{i=1}^n 2\pi r(x_i)\,h(x_i)\,\Delta x$

$\text{Vol} = \int_a^b 2\pi\cdot r(x_i)\cdot h(x_i)\,dx$

Volume of rotation

def: volume obtained by rotating region b/w curve & the axis of rotation



approx

Vol of slice $= \pi\cdot r^2(x)\,\Delta x$

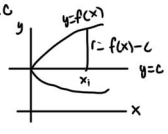
$= \int_a^b \pi\cdot r^2(x)\,dx$

$= \int_a^b \pi\cdot[f(x)]^2\,dx$ } exact

* if the axis of rotation is $y=c$

$\text{Vol} = \int_a^b \pi r^2\,dx$

$= \int_a^b \pi\,[F(x)-c]^2\,dx$



Math 1322

Lecture 4 - physics apps : work

$W = F \cdot d$ → m (mass: kg, distance: m, force: N, work: J or N·m)
↓ N
J or N·m

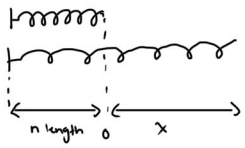
* if force is not fixed (varies on distance or smth else)

$\text{work} = \int_a^b f(x)\,dx$

Hooke's law

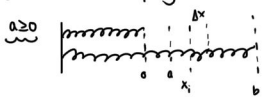
↳ force required to maintain a spring stretched x units beyond its natural length

$f(x) = kx$ (↳ spring constant)



$f(x) = kx$

so to stretch a spring from a to b

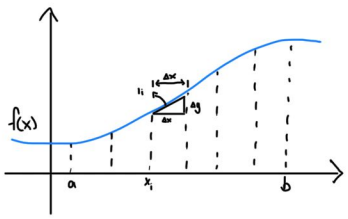


$w_i = f(x_i) \cdot \Delta x = k \cdot x_i \cdot \Delta x$

$w = \int_a^b kx\,dx$

Note: $\rho = \frac{\text{mass or weight}}{\text{vol or area or length}}$

Lecture 5 - average value; arc length



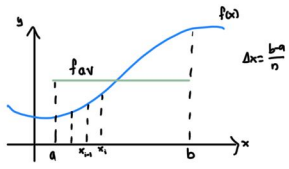
$\Delta x = \frac{b-a}{n}$

arc length

$\text{slope} = \frac{\Delta y}{\Delta x}$ l_i = length of tangent at x_i

exact length: $l = \int_{x=a}^{x=b} \sqrt{1+[f'(x)]^2}\,dx$

$l = \int_a^b \sqrt{(\frac{dx}{dt})^2+(\frac{dy}{dt})^2}\,dt$



$\Delta x = \frac{b-a}{n}$

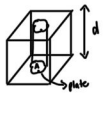
average value

exact $f_{av} = \frac{1}{b-a}\int_a^b f(x)\,dx$

Lecture 6 - physics apps: hydrostatic force; centre of mass

hydrostatic pressure

Pressure = $\frac{\text{force}}{\text{area}}$



→ weight of fluid (w) on this plate
$= \rho V g$ → gravity
↳ density ↳ Ad → distance/depth, area

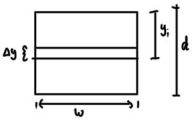
$w = \rho A d g$

→ pressure (P) on this plate
$= \frac{w}{A} = \frac{\rho A d g}{A} = \rho g d$ $\left(\frac{N}{m^2}\right)$ $\left(\frac{lb}{ft^2}\right)$

Hydrostatic pressure at depth $d = \rho g d$ (from the top of the water)

hydrostatic force on the sides

- hydrostatic pressure at depth y_i $= \rho \cdot g \cdot y_i$
- area of the strip $w\,\Delta y$



- hydrostatic force on this strip = pressure · area $= \rho g y_i w \Delta y$

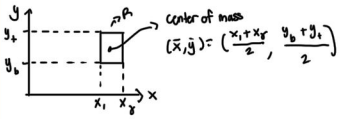
total approx hydrostatic force on this side $= \sum_{i=1}^{n} \rho \cdot g \cdot y_i \cdot w \cdot \Delta y$

total exact hydrostatic force on this side $= \lim_{n\to\infty} \sum_{i=1}^{n} \rho g y_i w \Delta y$
$= \int_0^d \rho \cdot g \cdot w \cdot y\,dy$ (↳ density, ↳ gravity, ↳ width, ↳ depth (d))

moment and center of mass

moment: the tendency to tip over or rotate around an axis

* Assume we have a rectangular plate of uniform density ρ is situated in the x-y plane



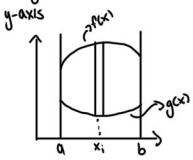
The mass of R = $\rho \cdot$ area

the moment of R about x-axis
M_x = mass · (signed) distance from center of mass to x-axis
$= \rho \cdot A \cdot \bar{y}$

The moment of R about y-axis
$M_y = \rho \cdot A \cdot \bar{x}$

* if the area is not regular (no formula)

The moment of the region R bounded by f(x), g(x), x=a, x=b about x-axis and y-axis



→ split the region into n rectangles each of width Δx
→ find the center of mass $\left(x_i, \frac{f(x_i)+g(x_i)}{2}\right)$

→ find moment of the rectangle about x-axis
$(M_x)_i = \rho \cdot A_i \cdot \text{distance}_x$
$= \rho \cdot (f(x_i) - g(x_i)) \cdot \Delta x \cdot \frac{f(x_i)+g(x_i)}{2}$

→ find approximate moment
approx $M_x = \sum_{i=1}^{n} (M_x)_i$

→ find exact moment
$M_x = \lim_{n\to\infty} \sum_{i=1}^{n} (M_x)_i$
$= \int_a^b \rho \cdot [f(x)-g(x)] \cdot \frac{[f(x)+g(x)]}{2}\,dx$
$= \frac{1}{2}\int_a^b \rho \cdot [f^2(x) - g^2(x)]\,dx$

→ find moment of the rectangle about y-axis
$(M_y)_i = \rho \cdot A_i \cdot \text{distance}_y$
$= \rho \cdot [f(x_i) - g(x_i)] \cdot \Delta x \cdot x_i$
approx $M_y = \sum_{i=1}^{n} \rho \cdot f(x_i) - g(x_i) \cdot \Delta x \cdot x_i$
$M_y = \int_a^b \rho \cdot x \cdot [f(x) - g(x)]\,dx$

Lecture 7 - differential equations; separable DEs; direction fields

Differential equations

$f(\frac{dy}{dx}, y, x) = 0$

an equation contain an independant and a dependant variables & a derivative

ex: $\frac{dy}{dx} = xy$ $x^2\frac{dy}{dx} = \sqrt{y^2+1}$ $v\frac{dv}{dt} = e^t$

→ classification of DES
order: the highest derivative in the DE
degree: the degree of the highest derivative

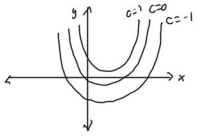
→ antiderivative are D.E. (1st order, 1st degree)

ex: $\frac{dy}{dx} = 3x - 4$
$dy = (3x-4)dx$
$\int dy = \int 3x - 4\,dx$
$y + c_1 = \frac{3}{2}x^2 - 4x + c_2$
$y = \frac{3}{2}x^2 - 4x + c$
equivalent to $\int 3x - 4\,dx$

Initial value problems (IVP):

The last example $\frac{dy}{dx} = 3x - 4$

↳ solution: $y = \frac{3}{2}x^2 - 4x + c$



→ this solution is called "general solution".
→ because of the constant c, this general solution is a family of curves with slope of tangent $3x - 4$
→ if we have a condition (initial conditional) i.e. $y(x_0) = y_0$, we can solve for c, and have "particular solution"
→ the initial condition specifies a point (x_0, y_0) on a particular curve.

Seperable D.E

→ finding an explicit solution is usually not possible. However, there are some "nice" D.E.s that we know how to find explicit solution for.

→ A seperable D.E is of the form $\frac{dy}{dx} = f(x) \cdot g(y)$

→ we can solve it by seperating x's from y's

$\frac{1}{g(y)}\,dy = f(x)\,dx \Rightarrow g(y) \neq 0$

$\int \frac{1}{g(y)}\,dy = \int f(x)\,dx$

Direction field

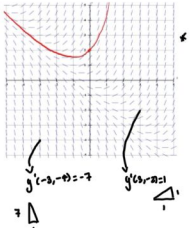
→ 1st order 1st degree D.E are slope of the tangent of the solution curve

→ for D.Es that we don't have an explicit formula of the solution, we can get a lot of information from slope

$y' = x + y$ (and) $y(0) = 2$

→ we cannot seperate variables ⇒ no explicit sol.
→ However, we can get the slope of the tangent at any point (x,y)

→ use I.C. to get information about sol



* the little tick marks are tangents to the general solution at this point

center of mass of R

we know that:
M_x = mass · (signed) distance from center of mass to x-axis
$= \text{mass} \cdot \bar{y}$
$M_y = \text{mass} \cdot \bar{x}$

mass $= \int \rho \cdot A$
$= \int_a^b \rho \int [f(x) - g(x)]\,dx$

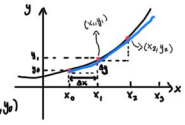
$\bar{x} = \frac{M_y}{\text{mass}}$, $\bar{y} = \frac{M_x}{\text{mass}}$

* if ρ is not fixed (depend on x and/or y) it has to be inside the integral
→ if fixed it can be extracted from integral

Lecture 8 - Euler's method; applications

Euler's method

- numerical method to approximate the solution of 1st order 1st degree DES
- we know that $y'(x,y)$ is the slope of the tangent to the solution curve at any point (x,y)
- Given an initial condition $y(x_0) = y_0$ we can use $y'(x)$, to get approximate values of the solution $y(x_i)$ at some selected points x_i
- $y(x)$ is the solution of I.V.P
- from x_0 take small step of size Δx $x_1 = x_0 + \Delta x$
- we know $f(x,y) = y' \approx \frac{\Delta y}{\Delta x}$



$\frac{\Delta y}{\Delta x} = f(x_0, y_0) \Rightarrow \frac{y_1 - y_0}{\Delta x} = f(x_0, y_0)$

$y_1 = y_0 + \Delta x\, f(x_0, y_0)$

- general formula of euler's using step size Δx

$x_i = x_{i-1} + \Delta x$
$y_i = y_{i-1} + \Delta x\, f(x_{i-1}, y_{i-1})$ → is a derivative

Apps of DEs

Newton's law of heating & cooling

$\frac{dT}{dt} = k(T - T_s)$ $T(t) = T_s + Ae^{kt}$

- The temperature (T) of an object increases (or decrease) at a rate proportional to the temperature difference between the object and the surroundings (T_s)
- The proportionality constant (k) depends on material

⇒ models for population growth

* exponential model:

- assumes population growth in ideal conditions
- population grows in a rate proportional to its size (P)

$\frac{dp}{dt} = kp$ (k → relative growth constant)
$\int \frac{1}{p}\,dp = \int k\,dt$
$\ln|p| = kt + c$
$p(t) = e^{c}e^{kt} = Ae^{kt}$

initial population is P_0
i.e. $p(0) = P_0$
$A = P_0$
$p(t) = P_0 \cdot e^{kt}$

* radioactive decay

→ the mass of a radioactive element decays exponentially

$\frac{dm}{dt} = km$ $m(t) = m_0 e^{kt}$

m_0 is the initial mass $m(0) = m_0$

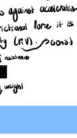
→ the half life of the element is the time required to reduce the present amount by half.

$m(t_{1/2}) = \frac{1}{2}m_0$
$m_0 e^{k \cdot t_{1/2}} = \frac{1}{2}m_0$
$kt_{1/2} = \ln\frac{1}{2}$
$k = \frac{\ln(1/2)}{t_{1/2}}$

$m(t) = m_0 e^{\frac{\ln(1/2)}{t_{1/2}}t} = m_0\left[e^{\ln(1/2)}\right]^{t/t_{1/2}}$
$= m_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}$

Free fall

if an object falls from height, air resistance works against acceleration since it is a frictional force it is proportional to velocity ($kv(t)$) → const



Newton Second law:
force = mass × acceleration
force = m·a
= weight - air resistance
$ma = mg - kv$
$m\frac{dv}{dt} = mg - kv$
$\frac{dv}{dt} = -\frac{k}{m}(v - \frac{mg}{k})$

$\int \frac{1}{v-\frac{mg}{k}}\,dv = \int -\frac{k}{m}\,dt$
integrate $\ln|v - \frac{mg}{k}| = -\frac{k}{m}t + c$
$v - \frac{mg}{k} = c_1 e^{-\frac{k}{m}t}$
$v(t) = \frac{mg}{k} + A \cdot e^{-\frac{k}{m}t}$
initial condition $v(0) = 0$ [initial velocity]
$0 = \frac{mg}{k} + A \Rightarrow A = -\frac{mg}{k}$
$v(t) = \frac{mg}{k}(1 - e^{-\frac{k}{m}t})$

Lecture 9 - sequences

→ An (infinite) sequence is an infinite list of numbers called elements or terms of the sequence.

$$\{a_1, a_2, a_3, \dots, a_n, a_{n+1}, \dots\}$$

$$\{a_n\}_{n=1}^{\infty} = \{a_n\}$$

→ they can be given in 3 ways

1. list few terms [Little pattern is obvious]
   ex: $\{1, 2, 3, \dots\}$
   $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$
2. Recurrence relation & starting value(s)
   ex: $a_1 = a_2 = 1$, $a_{n+2} = a_{n+1} + a_n$
   $\{1, 1, 2, 3, 5, 8, 13, \dots\}$
3. formula for a_n
   ex: $a_n = \frac{n^2}{n+1}$
   $\{\frac{1}{2}, \frac{4}{3}, \frac{9}{4}, \dots\}$

defs:

→ A sequence is bounded below by L, if $a_n \geq L$ for all n
→ A sequence is bounded above by M, if $a_n \leq M$ for all n
→ A sequence is bounded if it is bounded below & above

ex: $a_n = \frac{1}{n}$, $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ $L = 0$, $M = 1$

→ A sequence is increasing, if $a_{n+1} > a_n$ for all n
→ A sequence is decreasing, if $a_{n+1} < a_n$ for all n

→ A sequence is monotonic if it is increasing or decreasing
ex: $a_n = n+1$ increasing, $a_n = \frac{1}{n+2}$ decreasing

→ A sequence is alternating, if each pair of consecutive terms has opposite sign; a_n, a_{n+1}
ex: $a_n = (-1)^n$ $\{-1, 1, -1, \dots\}$

→ A sequence is convergent, if $\lim_{n\to\infty} a_n$ exists otherwise it is divergent.

limits rules

$\lim_{n\to\infty} a_n = L$, $\lim_{n\to\infty} b_n = M$

i) $\lim_{n\to\infty} (a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n = L \pm M$

ii) $\lim_{n\to\infty} (a_n b_n) = LM$

iii) $\lim_{n\to\infty} \left(\frac{a_n}{b_n}\right) = \frac{L}{M}$ $(M \neq 0)$

iv) $\lim_{n\to\infty} (a_n)^p = L^p$ $a_n > 0$, $p > 0$

→ A sequence is a list of numbers. If we add them up we have a series
→ An infinite series is a sum of an infinite number of numbers

$$a_1 + a_2 + a_3 + \dots + a_n + \dots$$

$$\sum_{i=1}^{\infty} a_i$$

* we cannot add infinite number of numbers.

$a_1 + a_2 + a_3 + a_4 + a_5 + \dots$

$S_1 = a_1$

$S_2 = a_1 + a_2$

$S_3 = a_1 + a_2 + a_3$

→ what we get is a sequence of partial sums of the series
→ the n-th element of this sequence is $S_n = \sum_{i=1}^{n} a_i = a_1 + a_2 + \dots + a_n$

$$\{S_n\} = \{S_1, S_2, \dots, S_n, \dots\}$$

→ if the sequence $\{S_n\}$ converges, then the series has a finite sum, and hence convergent

$$\sum_{i=1}^{\infty} a_i = \lim_{n\to\infty} \sum_{i=1}^{n} a_i \quad \text{if } \lim_{n\to\infty} S_n = L \Rightarrow \sum_{i=1}^{\infty} a_i = L$$

$$= \lim_{n\to\infty} S_n$$

Lecture 10 - series: telescopic, geometric, test for divergence

telescopic series

ex: $\sum_{n=1}^{\infty} \frac{1}{n^2+3n+2}$ converge or diverge?

$a_n = \frac{1}{n^2+3n+2} \Rightarrow S_n = \sum_{n=1}^{n} \frac{1}{n^2+3n+2}$

$S_n = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots + \frac{1}{n^2+3n+2}$

$a_n = \frac{1}{n^2+3n+2} = \frac{1}{(n+2)\cdot(n+1)}$

$\frac{1}{n^2+3n+2} = \frac{A}{n+2} + \frac{B}{n+1} = \frac{1}{n+1} - \frac{1}{n+2}$

$S_n = \sum_{n=1}^{n} \frac{1}{n+1} - \frac{1}{n+2}$

$= (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + (\frac{1}{4} - \frac{1}{5}) + \dots$

$(\frac{1}{n} - \frac{1}{n+1}) + (\frac{1}{n+1} - \frac{1}{n+2})$

$S_n = \frac{1}{2} - \frac{1}{n+2}$

$\lim_{n\to\infty} S_n = \lim_{n\to\infty} (\frac{1}{2} - \frac{1}{n+2}) = \frac{1}{2}$

$\sum_{n=1}^{\infty} \frac{1}{n^2+3n+2} = \lim_{n\to\infty} S_n = \frac{1}{2}$

Divergence test

if $\lim_{n\to\infty} a_n \neq 0 \rightarrow$ series diverges

* if a series diverges, it does not mean that $\lim_{n\to\infty} a_n \neq 0$

* you cannot conclude convergence from this test.

so $\lim_{n\to\infty} a_n = 0$ does not guarantee convergence

Harmonic series

$\sum_{n=1}^{\infty} \frac{1}{n}$

$\lim_{n\to\infty} \frac{1}{n} = 0$

$\{S_n\} = \{1, 1.5, 1.83, 2.08\}$

$S_2 = 1 + \frac{1}{2}$

$S_8 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}$

$> 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$

$= 1 + \frac{3}{2}$

$S_{2^n} > 1 + \frac{n}{2}$

$\lim_{n\to\infty} S_{2^n} > \lim_{n\to\infty} 1 + \frac{n}{2} = \infty$

if $\lim_{n\to\infty} S_{2^n} = \infty$ then $\lim_{n\to\infty} S_n = \infty$

so harmonic series diverge despite the fact $\lim_{n\to\infty} a_n = 0$

Geometric series

A series of the form $\sum_{n=1}^{\infty} a r^{n-1}$ $(a \neq 0, r$ constant$)$

$\sum_{n=1}^{\infty} a r^{n-1} = a + ar + ar^2 + ar^3 + \dots$

The nth partial sum

$S_n = a + ar + ar^2 + \dots + ar^{n-1}$

$rS_n = ar + ar^2 + ar^3 + \dots + ar^n$

$S_n - rS_n = a - ar^n$

$S_n(1-r) = a(1-r^n)$

$S_n = \frac{a(1-r^n)}{1-r}$ $r \neq 1$

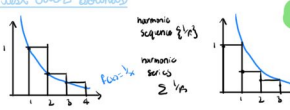
if $|r| < 1$ then $r^n \to 0$

$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r}$

Then $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$ converge

if $|r| \geq 1$ the geometric series diverge

Lecture 11 - integral test, remainder test, error bounds



$\sum_{n=1}^{\infty} \frac{1}{n}$ resembles $\int_1^{\infty} \frac{1}{x} dx$

$\sum_{n=1}^{\infty} \frac{1}{n} > \int_1^{\infty} \frac{1}{x} dx = \infty$

so the series diverges

$\sum_{n=1}^{\infty} \frac{1}{n} \leq 1 + \int_1^{\infty} \frac{1}{x} dx = \infty$

we can't conclude divergence from this.

Integral test

→ $f(x)$ is continuous, positive & decreasing on $[1, \infty)$, let $a_n = f(n)$

$\sum_{n=1}^{\infty} a_n$ is convergent if & only if $\int_1^{\infty} f(x) dx$ is convergent

ex: $\sum_{n=2}^{\infty} \frac{2}{n \cdot (\ln n)^3}$

$\rightarrow f(x)$ is continuous, positive & decreasing

$2 \int_2^{\infty} \frac{1}{n \cdot (\ln n)^3} dn$ $u = \ln n$, $\frac{du}{dn} = \frac{1}{n}$

$\int_{\ln 2}^{\infty} \frac{1}{u^3} du = -\frac{1}{2u^2} \Big|_{\ln 2}^{\infty} = -\frac{1}{2(\ln n)^2} \Big|_2^{\infty}$

$= -\frac{1}{2(\ln \infty)^2} + \frac{1}{2(\ln 2)^2} = \frac{1}{2(\ln 2)^2}$

p-series

$\sum_{n=1}^{\infty} \frac{1}{n^p}$

comparing integral: $p \neq 1$ converge $\int_1^{\infty} \frac{1}{x^p} dx$ $|p| > 1$ diverge

then $\sum \frac{1}{n^p}$ $|p| > 1$ converge $|p| \leq 1$ diverge

ex: $\sum_{n=1}^{\infty} \frac{4n}{n^2+8}$

$f(x) = \frac{4x}{x^2+8}$, $f'(x) = \frac{32 - 4x^2}{(x^2+8)^2}$

cont, + ve, decreasing

$\int \frac{4x}{x^2+8} dx = \infty$ (series diverges)

The comparison test

ex: $\sum_{n=1}^{\infty} \frac{4n+8}{n^2}$

→ this series looks like $\sum \frac{1}{n}$ (divergent)
so we suspect our series diverges
→ we will look for lower bound like $\sum \frac{1}{n}$ (or a multiple of)

we know that $4n+8 \geq 4n$ for all $n \geq 1$

then $\frac{4n+8}{n^2} \geq \frac{4n}{n^2} = \frac{4}{n}$

$\sum_{n=1}^{\infty} \frac{4n+8}{n^2} \geq 4 \sum_{n=1}^{\infty} \frac{1}{n} = \infty$ divergs

Suppose $\sum a_n$ & $\sum b_n$ are series with +ve terms

if $\sum b_n$ is convergent, & $a_n \leq b_n$ for all n then $\sum a_n$ is convergent
if $\sum b_n$ is divergent, & $a_n \geq b_n$ for all n then $\sum a_n$ is divergent

ex:

$\sum_{n=1}^{\infty} \frac{3}{n^2+3n+1}$

it looks like $\sum \frac{3}{n^2}$ (p-series $p > 1$ convergent)

$n^2 + 3n + 1 \geq n^2$ for all $n \geq 1$

then $\frac{3}{n^2+3n+1} \leq \frac{3}{n^2}$

$\sum_{n=1}^{\infty} \frac{3}{n^2+3n+1} \leq \sum_{n=1}^{\infty} \frac{3}{n^2}$ so convergent

Lecture 12 - comparison tests

limit comparison test

Suppose that $\sum a_n$, $\sum b_n$ are series with positive terms. if $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is a finite number and $c > 0$ then, both series behave in the same way both converge or both diverge.

ex: $\sum_{n=1}^{\infty} \frac{1}{3^n - 2}$

looks like $\sum \frac{1}{3^n}$ → geometric series $a = 1/3$, $r = 1/3$

* using limit comparison test

$\lim_{n\to\infty} \frac{1/3^n}{1/3^n - 2} = \lim_{n\to\infty} \frac{3^n - 2}{3^n}$

$= \lim_{n\to\infty} 1 - \frac{2}{3^n} = 1$

* comparison test leads us nowhere

$3^n - 2 < 3^n$ for $n \geq 1$

then $\sum_{n=1}^{\infty} \frac{1}{3^n - 2} > \sum_{n=1}^{\infty} \frac{1}{3^n}$

ex: $\sum_{n=1}^{\infty} \frac{2n+1}{\sqrt{n^3+1}}$

$\lim_{n\to\infty} \frac{\frac{2n+1}{\sqrt{n^3+1}}}{\frac{1}{\sqrt{n}}} = \lim_{n\to\infty} \frac{2n^{3/2} + n^{1/2}}{\sqrt{n^3+1}}$

$= \lim_{n\to\infty} \frac{2 + 1/n}{\sqrt{1 + 1/n^3}} = 2 > 0$

$\sum \frac{2n}{\sqrt{n^3}} = \sum \frac{1}{n^{1/2}}$

p-series with $p \leq 1$ diverges

$2n+1 > 2n$, $n^3+1 > n^3$ for all $n \geq 1$

→ comparison test leads us nowhere

the series diverges.

Error bounds for the integral test:

Suppose that we discovered that $\sum_{n=1}^{\infty} a_n$ is convergent using the integral test,

can we evaluate the exact value $\sum_{n=1}^{\infty} a_n$?

How close is the nth partial sum to the actual sum?

let $S = \sum_{n=1}^{\infty} a_n$

$S = a_1 + a_2 + a_3 + \dots + a_n + a_{n+1} + \dots$

$S = S_n + R_n$

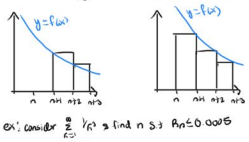
$= \sum_{n=1}^{n} a_n + \sum_{n=n+1}^{\infty} a_n$

S_n $\quad$ R_n

$R_n = S - S_n$ the difference between the actual sum & the nth partial sum

The remainder estimate of the integral test

Suppose that $a_n = f(n)$, where $f(x)$ is continuous, +ve, and decreasing for all $x \geq n$ and $\sum_{n=1}^{\infty} a_n$ is convergent then $\int_{n+1}^{\infty} f(x) \leq R_n \leq \int_n^{\infty} f(x)$



ex: consider $\sum_{n=1}^{\infty} 1/n^3$ & find n s.t $R_n \leq 0.0005$

$R_n = S - S_n \leq \int_n^{\infty} 1/x^3 dx$

$R_n \leq \lim_{t\to\infty} \int_n^t 1/x^3 dx$

$= \lim_{t\to\infty} \left[\frac{1}{-2x^2}\right]_n^t$

$= \lim_{t\to\infty} -\frac{1}{2t^2} + \frac{1}{2n^2}$

$= \frac{1}{2n^2} \leq 0.0005$

$n^2 \geq 1000$

$n \geq 31.62$

we want $n \geq 32$

32nd partial sum $S_{32} = \sum_{n=1}^{32} 1/n^3$

satisfies $R_n \leq 0.0005$

→ knowing R_n, can we estimate S?

we can get an estimate.

$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$

$\int_{n+1}^{\infty} f(x) dx \leq S - S_n \leq \int_n^{\infty} f(x) dx$

$S_n + \int_{n+1}^{\infty} f(x) dx \leq S \leq S_n + \int_n^{\infty} f(x) dx$

ex: lowere & upper bounds of $\sum_{n=1}^{\infty} 1/n^3$

by using the 10th partial sum

$S_{10} + \int_{10}^{\infty} 1/x^3 dx \leq S \leq S_{10} + \int_{10}^{\infty} 1/x^3 dx$

$S_{10} = 1 + \frac{1}{2^3} + \frac{1}{3^3} + \dots + \frac{1}{10^3} \approx 1.197$

$\int_{10}^{\infty} 1/x^3 dx \approx 0.004432$, $\int_{10}^{\infty} 1/x^3 dx = 0.005$

so, $1.197532 + 0.004432 \leq S \leq 1.197532 + 0.005$

$1.201964 \leq S \leq 1.202532$

Lecture 13 - alternating series (AST & ASET) absolute convergence

Alternating series

series in which opposite terms have opposite signs.

ex: $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$

ex: $\sum_{n=0}^{\infty} (-1)^n \sqrt{n} = 0 - 1 + \sqrt{2} - \sqrt{3} + 2 - \dots$

Suppose $\sum a_n$ is an alternating series.

if $b_n = |a_n|$ the nth term of the alternating series is

$a_n = (-1)^n b_n$ or $a_n = (-1)^{n-1} b_n$

Alternating series estimation theorem [A.S.E.T]

if $b_n > 0$, and $S = \sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is an alternating series that satisfies

i) $b_{n+1} \leq b_n$ ii) $\lim_{n\to\infty} b_n = 0$

then, $|R_n| = |S - S_n| \leq b_{n+1}$

i.e $-b_{n+1} \leq S - S_n \leq b_{n+1}$

$S_n - b_{n+1} \leq S \leq S_n + b_{n+1}$

ex: $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ How many terms do we need to $|error| = |R_n| < 0.01$

This is alternating series with $b_n = 1/n!$

i) $b_{n+1} = 1/(n+1)! = 1/(n(n+1)!)$

since $n! < n(n+1)!$

$\leq 1/n! = b_n$

ii) $\lim_{n\to\infty} b_n = \lim_{n\to\infty} 1/n! = 0$

The series is convergent by (A.S.T)

by A.S.E.T, $|R_n| \leq b_{n+1}$

if $b_{n+1} \leq 0.01$

$1/(n+1)! \leq 1/100$

$(n+1)! \geq 100$

$n \geq 4$ (since $3! < 100$, $4! > 100$)

ex: find the sum $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3}$ to within 0.005

i) $b_{n+1} = \frac{1}{(n+1)^3} < \frac{1}{n^3} = b_n$

ii) $\lim_{n\to\infty} \frac{1}{n^3} = 0$

convergent by A.S.T

$|R_n| \leq b_{n+1} \leq 0.005$

$\frac{1}{(n+1)^3} \leq 0.005$

$(n+1)^3 \geq 200$

$n+1 \geq 5.848$

$n \geq 4.848$

so $n \geq 5$

$S_5 - b_6 \leq S \leq S_5 + b_6$

Absolute V.S. conditional convergence

→ A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ is convergent

ex: $\sum \frac{(-1)^n}{n^2}$ is absolutely convergent

as $\sum \frac{1}{n^2}$ is convergent

Theorem: if a series $\sum a_n$ is absolutely convergent then it is convergent. $\sum |a_n|$ convergent $\Rightarrow \sum a_n$ convergent

Pf: the addition (subtraction) of convergent series is a convergent series.

$\sum a_n = \sum (a_n + |a_n|) - \sum |a_n|$ → convergent

we have $0 \leq a_n + |a_n| \leq 2|a_n|$

then $\sum (a_n + |a_n|) \leq \sum 2|a_n|$

then $\sum (a_n + |a_n|)$ is convergent by the comparison test

therefore $\sum a_n$ is convergent

→ A series is called conditionally convergent if $\sum a_n$ is convergent, but $\sum |a_n|$ is divergent

ex: $\sum \frac{(-1)^n \cdot n}{n^2+1}$ convergent or divergent? → which type?

$\lim_{n\to\infty} \frac{n}{n^2+1} = \frac{1/n}{1 + 1/n^2} = 0$ consider $\sum |a_n| = \sum \frac{n}{n^2+1}$

→ convergent → divergent

series is conditionally convergent.

ex: $\sum \frac{\sin(n)}{n^3}$ convergent or divergent. if convergent, which type?

→ not alternating

$\sum |a_n| = \sum \left|\frac{\sin(n)}{n^3}\right| \leq \sum \frac{1}{n^3}$ → comparison test → convergent

$\sum \frac{\sin(n)}{n^3}$ is absolutely convergent

$= \sum (\frac{1}{3})^{n-1}$ geometric $|r| = \frac{1}{3} < 1$

Lecture 14 - ratio & root test, series test strategy

Ratio test

consider $\sum a_n$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$ $\begin{cases} L < 1 & \text{(absolutely convergent)} \\ L > 1 \text{ or } \infty & \text{(divergent)} \\ L = 1 & \text{(inconclusive)} \end{cases}$

Ex: $\sum \frac{b^n}{n!}$

$\frac{b^{n+1}}{(n+1)!} \cdot \frac{n!}{b^n} = \frac{b^n \cdot b}{(n+1) \cdot n!} \cdot \frac{n!}{b^n}$

$\lim_{n\to\infty} \frac{b}{n+1} = 0 < 1$

abs convergent

Ex: $\sum \frac{(-1)^n 2^n}{n^2}$

$\frac{2^{n+1}}{(n+1)^2} \cdot \frac{n^2}{2^n} = \frac{2^n \cdot 2}{n^2+2n+1} \cdot \frac{n^2}{2^n}$

$\lim_{n\to\infty} \frac{2n^2}{n^2+2n+1} = 2 > 1$

divergent

Root test

consider $\sum a_n$

$\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ $\begin{cases} L < 1 & \text{(absolutely convergent)} \\ L > 1 \text{ or } \infty & \text{(divergent)} \\ L = 1 & \text{(inconclusive)} \end{cases}$

ex: $\sum \left(\frac{5n+3}{7n+4}\right)^n$

$\lim_{n\to\infty} \frac{5n+3}{7n+4} = \frac{5}{7} < 1$

absolutely convergent

ex: $\sum \left(\frac{-2n}{n+1}\right)^{5n}$

$\lim_{n\to\infty} \left|\left(\frac{-2n}{n+1}\right)^5\right|$

$= \lim_{n\to\infty} \left(\frac{2n}{n+1}\right)^5$ (limit theorem)

$= 2^5 > 1$

divergent

Lecture 15 - power series; power series representations

Power series:

⇒ a power series centered at $x=a$ has the form

$$\sum_{n=0}^{\infty} C_n(x-a)^n = C_0 + C_1(x-a) + C_2(x-a)^2 + \cdots$$

⇒ if we plug a particular value, say $x=b$, we get

$$\sum_{n=0}^{\infty} C_n(b-a)^n = C_0 + C_1(b-a) + C_2(b-a)^2 + \cdots$$

which is a numerical series $\sum_{n=0}^{\infty} a_n$

⇒ power series may be convergent for some values of x clearly for $x=a$ the series converges to C_0

$$\sum_{n=0}^{\infty} C_n(a-a)^n = C_0 + C_1(0) + C_2(0) + \cdots = C_0$$

Then all power series converge at the center

⇒ a power series centered at $a=0$ looks like

$$\sum_{n=0}^{\infty} C_n(x-0)^n = C_0 + C_1x + C_2x^2 + \cdots$$

⇒ a power series defines a function

$$f(x) = \sum_{n=0}^{\infty} C_n(x-a)^n$$

The domain all the values x for which the series converges

Radius & interval of convergence

for a power series $\sum_{n=0}^{\infty} C_n(x-a)^n$, there are 3 cases

1) The series only converges for $x=a$

2) The series converges for all $x \in \mathbb{R}$

3) There is a positive number R s.t. Series converges if $|x-a|<R$; Series converges if $|x-a|>R$

* R is called the radius of convergence (3a in case i), $R=\infty$ in case ii)

⇒ we use R to define the interval of convergence (I)

the series converges for $|x-a|<R$

i.e. $a-R<x<a+R$

& we have to discuss the point $x=a-R$, $x=a+R$

→ The interval of convergence I could be

$a-R<x<a+R$ ⇒ $|x-a|<R$

or $a-R \le x < a+R$ [ )

or $a-R < x \le a+R$ ( ]

or $a-R \le x \le a+R$ [ ] ⇒ $|x-a| \le R$

Therfore, ⇒ $I=\{a\}$ in the interval of convergence in case i)

⇒ $I=(-\infty,\infty)$ is the interval of convergence in case ii)

ex: find the radius & interval of convergence of $\sum_{n=1}^{\infty} \frac{\ln n}{n}(x+1)^n$

Ratio test $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$

$$= \lim_{n\to\infty}\left|\frac{\ln(n+1)}{n+1}(x+1)^{n+1} \cdot \frac{n}{\ln(n)(x+1)^n}\right|$$

$$= \lim_{n\to\infty}\frac{n\ln(n+1)}{(n+1)\ln(n)}|x+1| = \lim_{n\to\infty}\frac{n}{n+1}\lim_{n\to\infty}\frac{\ln(n+1)}{\ln(n)}|x+1| = |x+1|$$

⇒ the series converges for $|x+1|<1$ ⇒ $-2<x<0$

radius of convergence $R=1$

at $x=-2$

$$\sum_{n=1}^{\infty}\frac{\ln(n)}{n}(x+1)^n = \sum_{n=1}^{\infty}(-1)^n\frac{\ln(n)}{n}$$

converge A.S.T (check)

at $x=0$

$$\sum_{n=0}^{\infty}\frac{\ln(n)}{n}(x+1)^n = \sum_{n=1}^{\infty}\frac{\ln(n)}{n}$$

diverge (compare with $\sum 1/n$ or integral test)

the interval of convergence I is $-2 \le x < 0$ or $[-2,0)$

ex: power series, coeff $C_n=1$ for all n & centered at $a=0$. what series looks like & what is domain

$$\sum_{n=0}^{\infty}1\cdot(x-0)^n = \sum_{n=0}^{\infty}x^n = 1+x+x^2+x^3+x^4+\cdots$$

e.g. $P(1/2) = (1/2)^0+(1/2)^1+(1/2)^2+(1/2)^3+\cdots = \sum_{n=0}^{\infty}(1/2)^n$

geometric series with $|r|=1/2$ converges to $\frac{a}{1-r} = \frac{1}{1-1/2} = 2$

but $P(2)$ is geometric series with $|r|=2>1$ diverges.

⇒ the series (function) $P(x)$ is defined for all $|x|<1$ {the values for which the geometric series converges} & it converges to $\frac{1}{1-x}$.

$$P(x) = \sum_{n=0}^{\infty}x^n \begin{cases} \text{converges to } \frac{1}{1-x} & \text{if } |x|<1 \\ \text{diverges} & \text{if } |x|\ge 1 \end{cases}$$

the domain is all x s.t. $|x|<1$ ⇒ $-1<x<1$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty}x^n = 1+x+x^2+x^3+\cdots$$

⇒ we can use the ratio test to find values of x for which the series is convergent.

ex: Find values of x for which the power series $\sum_{n=1}^{\infty}\frac{x^n}{n^2 2^n}$ is convergent ($a=0$)

Use the ratio test $a_n = \frac{x^n}{n^2 2^n}$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^{n+1}}{(n+1)^2 2^{n+1}}\cdot\frac{n^2 2^n}{x^n}\right| = \lim_{n\to\infty}\left|\frac{x\cdot n^2}{(n+1)^2\cdot 2}\right| = \lim_{n\to\infty}\left|\frac{x}{2}\cdot\frac{n^2}{n^2+2n+1}\right| = \left|\frac{x}{2}\right|$$

⇒ the ratio test tells us that the series is convergent if $\left|\frac{x}{2}\right|<1$ ⇒ $|x|<2$ ⇒ $-2<x<2$ & divergent if $\left|\frac{x}{2}\right|>1$ ⇒ $|x|>2$ ⇒ $x>2$ & $x<-2$

⇒ the test is inconclusive if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$

* radius of convergence is $R=2$

• if $x=-2$

then $\sum_{n=0}^{\infty}\frac{x^n}{n^2 2^n} = \sum_{n=0}^{\infty}\frac{(-2)^n}{n^2\cdot 2^n} = \sum_{n=0}^{\infty}\frac{(-1)^n\cdot 2^n}{n^2\cdot 2^n} = \sum_{n=0}^{\infty}\frac{(-1)^n}{n^2}$

converges by A.S.T.

• if $x=2$

then $\sum_{n=0}^{\infty}\frac{x^n}{n^2 2^n} = \sum_{n=0}^{\infty}\frac{2^n}{n^2 2^n} = \sum_{n=0}^{\infty}\frac{1}{n^2}$

p-series $p>1$ converges

The power series converges for all $-2 \le x \le 2$ or $|x| \le 2$ or $[-2,2]$

ex: $\sum_{n=1}^{\infty}\frac{n!\,x^n}{n^3}$

$$\lim_{n\to\infty}\left|\frac{(n+1)!\,x^{n+1}}{(n+1)^3}\cdot\frac{n^3}{n!\,x^n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)\cdot x\cdot n^3}{(n+1)^3}\right| = \lim_{n\to\infty}\frac{n^3+n^2}{(n+1)^3}|x| = \infty$$

The series diverge for all $x \in \mathbb{R}$ except at the center $x=0$. ∴ $R=0$ & the interval of convergence $I=\{0\}$

ex: $\sum_{n=0}^{\infty}\frac{(-1)^n\cdot x^{2n}}{(2n)!}$

$$\lim_{n\to\infty}\left|\frac{(-1)^{n+1}\cdot x^{2n+2}}{(2n+2)!}\cdot\frac{(2n)!}{(-1)^n\cdot x^{2n}}\right|$$

$$\lim_{n\to\infty}\left|\frac{(-1)\cdot x^2}{(2n+2)\cdot(2n+1)}\right|$$

$$\lim_{n\to\infty}\frac{-1}{(2n+2)(2n+1)}x^2 = 0 < 1$$

The series is convergent for all x

$R=\infty$ & $I=(-\infty,\infty)$

Lecture 15

Representation of a function as a power series

⇒ power series defines a function $f(x) = \sum_{n=0}^{\infty}C_n(x-a)^n$ whose domain is the interval of convergence.

⇒ given a function we can represent it as a power series

ex: $f(x) = \frac{1}{1-x}$ (geometric series)

$f(x) = \sum_{n=0}^{\infty}x^n = 1+x+x^2+\cdots$ for $|x|<1$

Power series representation for $f(x) = \frac{1}{1-x}$ is $f(x) = \sum_{n=0}^{\infty}x^n$

interval of convergence: $-1<x<1$

radius of convergence: $R=1$

ex: $g(x) = \frac{1}{1+x}$

$f(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty}x^n$ for $|x|<1$

& $g(x) = f(-x)$ then $g(x) = \sum_{n=0}^{\infty}(-x)^n$

P.S.S. for $g(x) = \frac{1}{1+x}$ is $g(x) = \sum_{n=0}^{\infty}(-1)^n\cdot x^n = \sum_{n=0}^{\infty}(-1)^n\cdot(x)^n$

interval of convergence: $-1<x<1$ ⇒ for $|-1|<1$ ⇒ for $|x|<1$

ex: $h(x) = \frac{1}{x+3}$

$$\frac{1}{3+x} = \frac{1}{3(1+x/3)} = \frac{1}{3}\left(\frac{1}{1-(-x/3)}\right) \quad r=|-x/3|$$

$$\frac{1}{1-r} = \sum_{n=0}^{\infty}[r(x)]^n$$

$$\frac{1}{3}\sum_{n=0}^{\infty}\left[-\frac{x}{3}\right]^n = \sum_{n=0}^{\infty}\frac{1}{3}(-1)^n\cdot\left(\frac{x^n}{3^n}\right) = \sum_{n=0}^{\infty}\frac{(-1)^n x^n}{3^{n+1}}$$

interval of convergence: $(-3,3)$ for $|-x/3|<1$ ⇒ $|x|<3$

ex: $p(x) = \frac{x^2}{8+2x} = \frac{x^2}{8}\left(\frac{1}{1+x/4}\right) = \frac{x^2}{8}\left(\frac{1}{1-(-x/4)}\right)$

$$p(x) = \frac{x^2}{8}\sum_{n=0}^{\infty}\frac{(-1)^n x^n}{4^n} = \sum_{n=0}^{\infty}\frac{(-1)^n x^{n+2}}{8\cdot 4^n}$$

interval of convergence is $|-x/4|<1$ ⇒ $-4<x<4$

Differentiation & integration

⇒ The power series representation of a function are (infinite) polynomials are "nice" functions that are easy to integrate & differentiate.

⇒ if the power series $\sum_{n=0}^{\infty}C_n(x-a)^n$ has radius of convergence R_1 and we can represent the function $f(x)$ as $f(x) = \sum_{n=0}^{\infty}C_n(x-a)^n$, then

i) $f'(x) = \frac{d}{dx}\left[\sum_{n=0}^{\infty}C_n(x-a)^n\right] = \sum_{n=0}^{\infty}C_n\frac{d}{dx}(x-a)^n = \sum_{n=0}^{\infty}C_n\cdot n(x-a)^{n-1}$

ii) $\int f(x)dx = \int\sum_{n=0}^{\infty}C_n(x-a)^n = C + \sum_{n=0}^{\infty}C_n\frac{(x-a)^{n+1}}{n+1}$

* the radius of convergence for both power series $f'(x)$ & $\int f(x)dx$ are both still R but the interval of convergence are not necessarily the same. → during integration we might gain convergence at an end point. during differentiation, we might lose convergence at an end point.

ex: $f(x) = \tan^{-1}(x)$

we know that $\tan^{-1}(x) = \int\frac{1}{1+x^2}dx$

we have $\frac{1}{1+x} = \sum_{n=0}^{\infty}(-1)^n x^n$ on $|x|<1$

then $\frac{1}{1+x^2} = \sum_{n=0}^{\infty}C_n x^{2n}$ on $|-x^2|<1$

$$\tan^{-1}(x) = \int\frac{1}{1+x^2}dx = \int\sum_{n=0}^{\infty}C_n x^{2n}dx = C + \sum_{n=0}^{\infty}C_n\frac{x^{2n+1}}{2n+1}$$

$$= C + \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1} \quad \text{on } |x^2|<1$$

$x=1$: $\sum_{n=0}^{\infty}\frac{(-1)^n(1)^{2n+1}}{2n+1}$ converges by A.S.T (general p-series)

then $\ln(1+x) = \int\frac{1}{1+x}dx = \int\sum_{n=0}^{\infty}(-1)^n x^n dx = C + \sum_{n=0}^{\infty}(-1)^n\frac{x^{n+1}}{n+1}$ on $-1<x\le 1$, $(-1,1]$

solve for C: let $x=0$ ⇒ $\ln(1+0) = C = 0$

$\ln(1+x) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{n+1}}{n+1}$ on $|x|<1$ & $-1<x\le 1$

$x=-1$: $\sum_{n=0}^{\infty}\frac{(-1)^n(-1)^{n+1}}{n+1} = \sum_{n=0}^{\infty}\frac{(-1)^{2n+1}}{n+1} = \sum_{n=0}^{\infty}\frac{-1}{n+1} = -\sum_{n=0}^{\infty}\frac{1}{n+1}$ diverge by integral test

$$\tan^{-1}(x) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1}$$

check for the end points

at $x=1$: $\sum_{n=0}^{\infty}\frac{(-1)^n(1)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty}\frac{(-1)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty}\frac{-1}{2n+1}$ converges by A.S.T

at $x=-1$: $\sum_{n=0}^{\infty}\frac{(-1)^n(-1)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty}\frac{(-1)^{3n+1}}{2n+1} = \sum_{n=0}^{\infty}\frac{1}{2n+1}$ converges by A.S.T

$$\tan^{-1}(x) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1} \quad \text{on } |x|\le 1$$

ex: $f(x) = \ln(2-x)$

$\ln(2-x) = \int\frac{-1}{2-x}dx$

$\ln(2-x) = \int\sum_{n=0}^{\infty}\frac{1}{2}\left(\frac{x}{2}\right)^n dx = \int\sum_{n=0}^{\infty}\frac{1}{2}\cdot\frac{x^n}{2^n}dx = \int\sum_{n=0}^{\infty}\frac{1}{2^{n+1}}x^n dx$

$= C + \sum_{n=0}^{\infty}\frac{x^{n+1}}{2^{n+1}(n+1)}$

$\ln(2-0) = C + 0 \Rightarrow C = \ln 2$

$\ln(2-x) = \ln 2 + \sum_{n=0}^{\infty}\frac{x^{n+1}}{2^{n+1}(n+1)}$ on $|x|<2$

to get $\ln$ for $x=0 \Rightarrow \ln(1+0)$

$\ln(2-x) = \ln 2 - \sum_{n=0}^{\infty}\frac{x^{n+1}}{2^{n+1}(n+1)}$ on $|x|<2$

$\ln 2 - \ln 2 = 0 = R_1 - R_2$

$R_1 = R_2$

$I = [-2,2)$

ex: $\frac{1}{(1+3x)^2}$

we know that $\frac{1}{1-x} = \sum_{n=0}^{\infty}x^n$ on $|x|<1$

then $\frac{1}{(1+x)^2} = \frac{d}{dx}\left[\frac{1}{1+x}\right] = \frac{d}{dx}\left[\sum_{n=0}^{\infty}(-1)^n x^n\right] = \sum_{n=0}^{\infty}(-1)^n n x^{n-1}$ on $|x|<1$

$$= \frac{1}{3}\sum_{n=0}^{\infty}(-1)^n n(3x)^{n-1} = \sum_{n=0}^{\infty}(-1)^n n 3^{n-1}x^{n-1}$$

on $|3x|<1$ or $-1<3x<1$

$$= \sum_{n=0}^{\infty}(-1)^{n+1}(n+1)3^n x^n \quad \text{on } |x|<\tfrac{1}{3}$$

two common ways you'll see from differentiation

ex: approximate $\int_0^{1/2}e^{-x^2}dx$ within an error of 0.001

we know that $e^x = \sum_{n=0}^{\infty}\frac{x^n}{n!}$

then $e^{-x^2} = \sum_{n=0}^{\infty}\frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}$

$$\int_0^{1/2}e^{-x^2}dx = \int_0^{1/2}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}dx = \left[\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)n!}\right]_0^{1/2} = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)2^{n+1}n!}$$

i) $\lim_{n\to\infty}b_n = 0$

ii) $b_{n+1} \le b_n$

so we can choose n for desired accuracy we want. if we want the error to be less than 0.1 we pick $b_{n+1} \le 0.1$

$\frac{1}{4\cdot 2^{n+1}} \le 0.1$

$n \ge 2.16$

so we can choose $n=3$ for desired accuracy we want. if we want the error to be less than 0.1 we pick $b_{n+1} \le 0.001$

$\frac{1}{4\cdot 2^{n+1}} \le 0.001$

$n \ge 7.64$

Lecture 16 - Taylor & Maclaurin series

Suppose that we have a power series representation for $f(x)$ on its interval of convergence centered around a.

$$f(x) = \sum_{n=0}^{\infty}C_n(x-a)^n = C_0 + C_1(x-a) + C_2(x-a)^2 + \cdots$$

$$C_0 = f(a)$$

$$C_1 = f'(a)$$

$$C_2 = \frac{f''(a)}{2!}$$

$$C_3 = \frac{f'''(a)}{3!}$$

$$C_n = \frac{f^{(n)}(a)}{n!}$$

notice $f(a) = C_0 + C_1(a-a) + C_2(a-a)^2 + \cdots = C_0$

$f'(x) = C_1 + 2C_2(x-a) + 3C_3(x-a)^2 + \cdots$

$f'(a) = C_1$

$f''(x) = 2C_2 + 2\cdot 3C_3(x-a) + \cdots$

$f''(a) = 2C_2 \Rightarrow C_2 = \frac{f''(a)}{2!}$

$f'''(x) = 3\cdot 2C_3 + \cdots$

$f'''(a) = 3!C_3 \Rightarrow C_3 = \frac{f'''(a)}{3!}$

in general $C_n = \frac{f^{(n)}(a)}{n!}$

the coefficients are determined by $f(x)$ & the center a means they are unique to f

$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$$

⇒ this is called the taylor series of f centered at a

⇒ in the case $a=0$, it is called Maclaurin series of f

$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n$$

ex: Maclaurin series for $f(x) = \frac{1}{1-x}$?

P.S.S. of $\frac{1}{1-x}$ centered at $a=0$ for $|x|<1$ is $\frac{1}{1-x} = \sum_{n=0}^{\infty}x^n$

& Maclaurin series for $\frac{1}{1-x}$ must coincide! So $\frac{f^{(n)}(0)}{n!} = 1$

$f(0) = 1 \Rightarrow C_0 = 1$

$f'(0) = 1 \Rightarrow C_1 = 1$

$f''(0) = 2 \Rightarrow C_2 = 1$

$f'''(0) = 6 \Rightarrow C_3 = 1$

$C_n = \frac{f^{(n)}(0)}{n!} = 1$

ex: Maclaurin series for $f(x) = e^x$

$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n$

$f(0) = e^0 \Rightarrow f(0) = 1$ $\quad f'(x) = e^x \Rightarrow f'(0) = 1$

$f''(x) = e^x \Rightarrow f''(0) = 1$ $\quad f^{(n)}(x) = e^x \Rightarrow f^{(n)}(0) = 1$

$e^x = \sum_{n=0}^{\infty}\frac{1}{n!}x^n$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

ratio test: $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^{n+1}}{(n+1)!}\cdot\frac{n!}{x^n}\right| = \lim_{n\to\infty}\frac{|x|}{n+1} = 0 < 1$ for all $x \in \mathbb{R}$

so $R=\infty$ & $I=(-\infty,\infty)$

ex: Use Maclaurin series to estimate $\int_0^1 e^{-x^2}dx$ within an error of 0.001

$$e^{-x^2} = \sum_{n=0}^{\infty}\frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}$$

$$\int_0^1 e^{-x^2}dx = \int_0^1\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}dx = \left[\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)n!}\right]_0^1 = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)n!}$$

by A.S.E.T remark: $|S-S_n| \le b_{n+1}$

$b_{n+1} \le 0.001$

ex: Determine Maclaurin series for $f(x) = \sin(x)$

$f(x) = \sin(x) \Rightarrow f(0) = 0$ $\quad f'(x) = \cos(x) \Rightarrow f'(0) = 1$

$f''(x) = -\sin(x) \Rightarrow f''(0) = 0$ $\quad f'''(x) = -\cos(x) \Rightarrow f'''(0) = -1$

$\sin(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n$

$= 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} + 0 + \cdots$

$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

$= \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!}$

ratio test: $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{x^{2n+3}}{(2n+3)!}\cdot\frac{(2n+1)!}{x^{2n+1}}\right| = 0 < 1$ for all $x \in \mathbb{R}$

$R=\infty$, $I=(-\infty,\infty)$

Lecture 17 - surfaces; functions of several variables

Functions of several variables

⇒ A function of (two) variables is a rule that assigns a real number for each ordered pair (x,y) in a set D.

$z = f(x,y)$

ex: $z = f(x,y) = x+y^2$

$y = f(x,y,z) = x^2+y^2+z^2$

several variables → f → real values

independent variables — dependent variable

⇒ Domain D: All values the independent variable can take so that the dependent variable is a real number

⇒ Range: All the values that $z = f(x,y)$ can take.

ex: find the domain of the function $z = \frac{\sqrt{x+y+1}}{x-1}$

we need to include all values of x & y that makes z a real number

$z = f(x,y)$ is real if $\begin{cases} x-1>0 \Rightarrow x>1 \\ x+y+1 \ge 0 \Rightarrow x+y \ge -1 \end{cases}$

D: $\{(x,y) \in \mathbb{R}^2 : x+y \ge -1 \text{ & } x>1\}$

⇒ the graph of the domain

we know $x+y \ge -1$ that is $y \ge -x-1$

so y is above the line $y=-x-1$ (including the line) & exclude the value $x=1$

ex: find the domain of $f(x,y) = \ln(1-y^2-x^2)$

Solution the domain

$f(x,y)$ is real if $1-y^2-x^2 > 0 \Rightarrow x^2+y^2 < 1$

D: $\{(x,y) \in \mathbb{R}^2 : x^2+y^2 < 1\}$

Solution: the function is defined for all the value inside the ellipse.... all the value inside the ellipse $x^2+y^2=1$ (ellipse is not included)

all the value inside the ellipse $x^2+y^2=1$ (ellipse is not included)

when $x=0 \Rightarrow y = \pm\frac{1}{2}$

when $y=0 \Rightarrow x = \pm 1$

the graph of a 3d function

if we have a function $z = f(x,y)$, then the graph of the function is all the triples (x,y,z) that satisfies the function $z = f(x,y)$.

→ in general the graph of $z = f(x,y)$ is a surface in the 3d plane.

→ to draw the graph of the function, we may use the traces or cross-section in which one of the variables is fixed & the others vary

→ a trace is the intersection between the surface & a plane parallel to a coordinate plane (like xy-plane).

ex: $y^2+x^2 = z^2$

$z = \pm\sqrt{x^2+y^2}$

the range is $\mathbb{R}$

ex: Sketch the traces of the function $z = x^2+y^2$

domain of function is $\mathbb{R}^2$

the range is $z \ge 0$ since $x^2+y^2 \ge 0$

& $z=0$ if $x=0$ & $y=0$

⇒ in xz plane ($y=0$) we have $z=x^2$ this is a parabola

⇒ in the plane ($z=1$) is a graph for $x^2+y^2=1$ this is a circle for $r=1$ with center $(0,0)$ & radius $r=1$, $x^2+y^2=(1)^2$

⇒ in the plane ($z=2$) is a graph for $x^2+y^2=2$ this is a circle for $r=\sqrt{2}$ with center $(0,0)$ & radius $r=\sqrt{2}$, $x^2+y^2=(\sqrt{2})^2$

Putting everything together, we get

ex: Sketch the traces of the function $z = \sqrt{x^2+y^2}$

the domain of the function is $\mathbb{R}^2$ & the range is $z \ge 0$

in the xz plane ($y=0$) $z = \sqrt{x^2} = |x|$

in the yz plane ($x=0$) $z = \sqrt{y^2} = |y|$

in the plane ($z=3$) the function $\sqrt{x^2+y^2} = 3$ ⇒ $x^2+y^2 = 9$ is a circle for $r=3$ & is a circle with center $(0,0)$ & radius $r=3$

Putting all together

ex: Sketch the traces of $4x^2+9y^2-9z^2=0$

we can write it as $\frac{x^2}{9}+\frac{y^2}{4}=z^2$, which is an ellipse in 2d

But we are in 3d, so z is appearing in the equation means that z can be anything. So what we have is an elliptical cylinder

ex: Maclaurin for $f(x) = \cos(x)$

$\cos(x) = \frac{d}{dx}[\sin(x)] = \frac{d}{dx}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$= \sum_{n=0}^{\infty}\frac{(-1)^n\cdot 2n\cdot x^{2n}}{(2n)!}$

$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

Binomial Series

Maclaurin series for $f(x) = (1+x)^k$

$f(0) = 1 \Rightarrow C_0 = 1$

$f(x) = (1+x)^5 \Rightarrow f(0) = 5$

$f'(x) = 5\cdot(1+x)^{5-1} \Rightarrow f'(0) = 5$

$f''(x) = 5\cdot(5-1)(1+x)^{5-2} \Rightarrow f''(0) = 5\cdot(5-1)$

$f'''(x) = 5\cdot(5-1)\cdot(5-2)$

$\vdots$

$f^{(n)}(x) = 5\cdot(5-1)\cdots(5-n+1)$

$(1+x)^5 = 1 + 5x + \frac{5\cdot(5-1)}{2!}x^2 + \frac{5\cdot(5-1)\cdot(5-2)}{3!}x^3 + \cdots$

this is called binomial series

* for $k \in \mathbb{R}$ & $n \in \mathbb{Z}$ s.t. $n \ge 3$, we denote by $\binom{k}{n}$ the binomial coefficient:

$\binom{k}{n} = \frac{k\cdot(k-1)\cdots(k-n+1)}{n!}$

and define $\binom{k}{0} = 1$, $\binom{k}{1} = k$, $\binom{k}{2} = \frac{k\cdot(k-1)}{2}$

* the binomial series can written as $(1+x)^k = \sum_{n=0}^{\infty}\binom{k}{n}x^n$

⇒ the ratio test: $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{k-n}{n+1}\cdot x\right| = |x|$

The binomial series converges for $|x|<1$ & $R=1$

& diverges for $|x|>1$.

ex: Find Maclaurin series for $\sqrt{1+x}$

$\sqrt{1+x} = (1+x)^{1/2} \Rightarrow k = 1/2$

$\binom{1/2}{n} = \frac{(1/2)\cdot(1/2-1)\cdot(1/2-2)\cdots(1/2-n+1)}{n!}$

$= \frac{(1/2)\cdot(-1/2)\cdot(-3/2)\cdots(3-2n)/2}{n!}$

$= \frac{(-1)^{n-1}\cdot 1\cdot 3\cdot 5\cdots(2n-3)}{2^n\cdot n!}$

$\binom{1/2}{0} = 1$, $\binom{1/2}{1} = 1/2$, $\binom{1/2}{2} = \frac{1/2\cdot(-1/2)}{2!} = -1/8$

$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \sum_{n=3}^{\infty}\frac{(-1)^{n-1}\cdot(1)\cdot(3)\cdots(2n-3)}{2^n\cdot n!}\cdot x^n$

ex: Maclaurin for $\frac{1}{(2+x)^3}$

$\frac{1}{(2+x)^3} = (2+x)^{-3} = [2(1+x/2)]^{-3}$

$= 2^{-3}(1+x/2)^{-3}$

$(1+x/2)^{-3} = \sum_{n=0}^{\infty}\binom{-3}{n}\left(\frac{x}{2}\right)^n$

$\binom{-3}{n} = \frac{(-3)\cdot(-3-1)\cdots(-3-n+1)}{n!} = \frac{(-3)\cdot(-4)\cdots(-n-2)}{n!}$

$= \frac{(-1)^n\cdot 3\cdot 4\cdots(n+2)}{1\cdot 2\cdot 3\cdots n} = \frac{(-1)^n(n+1)(n+2)}{2}$ (this holds for $n \ge 2$)

$\frac{1}{(2+x)^3} = \frac{1}{8}(1+x/2)^{-3} = \frac{1}{8}\sum_{n=0}^{\infty}\binom{-3}{n}\left(\frac{x}{2}\right)^n$

$= \frac{1}{8}\left[1 - 3\cdot\frac{x}{2} + \sum_{n=2}^{\infty}\frac{(-1)^n(n+1)(n+2)}{2}\cdot\left(\frac{x}{2}\right)^n\right]$

$= \frac{1}{8}\left[1 - \frac{3}{2}x + \sum_{n=2}^{\infty}\frac{(-1)^n(n+1)(n+2)}{2^n}\cdot x^n\right]$

Level curves

ex: draw a contour diagram for $f(x,y) = 6 - 3x - 2y$ using level curves for the values $k = -6, 0, 6, 12$

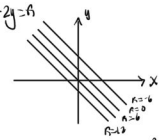
$6 - 3x - 2y = k$

$k=-6 \Rightarrow 3x + 2y = 12$

$k=0 \Rightarrow 3x + 2y = 6$

$k=6 \Rightarrow 3x + 2y = 0$

$k=12 \Rightarrow 3x + 2y = -6$



ex: draw the contour diagram for $f(x,y) = 2x^2 + y^2$ using $k = 1, 2, 4$

$2x^2 + y^2 = k$ ellipse equation



$k=1$ $2x^2 + y^2 = 1$
$x=0$ $y = \pm 1$
$y=0$ $x = \pm 1/\sqrt{2}$

$k=2$ $2x^2 + y^2 = 2$
$x=0$ $y = \pm\sqrt{2}$
$y=0$ $x = \pm 1$

$k=4$ $2x^2 + y^2 = 4$
$x=0$ $y = \pm 2$
$y=0$ $x = \pm\sqrt{2}$

ex: Draw level curves for $f(x,y) = y^2 - x^2$

$k = -2, -1, 0, 1, 2$

$y^2 - x^2 = k$ hyperbola

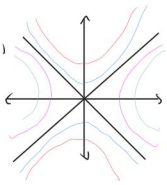
$k=-2$ $y^2 - x^2 = -2$
$\frac{x^2}{2} - \frac{y^2}{2} = 1$

$k=-1$ $x^2 - y^2 = 1$ $\Rightarrow$ $y^2 - x^2 = -1$

$k=0$ $y^2 - x^2 = 0$
$y = \pm x$

$k=1$ $y^2 - x^2 = 1$

$k=2$ $y^2 - x^2 = 2$
$\frac{y^2}{2} - \frac{x^2}{2} = 1$



Partial derivatives

$$y' = \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

Notations

$z = f(x,y)$

$z_x = f_x(x,y) = \frac{\partial z}{\partial x} = \frac{\partial f}{\partial x}$

at (a,b)

$z_x(a,b) = f_x(a,b) = \left.\frac{\partial z}{\partial x}\right|_{(a,b)} = \left.\frac{\partial f}{\partial x}\right|_{(a,b)}$

same for y

$z_y = f_y = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y}$

higher order partial derivative

$f'(x) = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$

$\frac{\partial}{\partial x}\left[\frac{\partial z}{\partial x}\right] = \frac{\partial^2 z}{\partial x^2} = (z_x)_x = z_{xx}$

$\frac{\partial}{\partial y}\left[\frac{\partial z}{\partial y}\right] = \frac{\partial^2 z}{\partial y^2} = (z_y)_y = z_{yy}$

$\frac{\partial}{\partial x}\left[\frac{\partial z}{\partial y}\right] = \frac{\partial^2 z}{\partial x \partial y} = (z_y)_x = z_{yx}$

$\frac{\partial}{\partial y}\left[\frac{\partial z}{\partial x}\right] = \frac{\partial^2 z}{\partial y \partial x} = (z_x)_y = z_{xy}$

Clairaut's theorem

f is defined on a disk D that contains the point (a,b). if partial derivatives f_{xy} & f_{yx} are both continuous on D, then $f_{xy} = f_{yx}$

tangent plane

equation of tangent line:

$\frac{y - y_0}{x - x_0} \approx f'(x_0)$

$y = f(x_0) + f'(x_0)(x - x_0)$



equation of the plane at point (x_0, y_0, z_0)

$z = f(x_0,y_0) + f_x(x_0,y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

linearization

approximate $z = f(x,y)$ near (x_0, y_0)

$f(x,y) \approx L(x,y)$

where

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x - x_0) + f_y(x_0,y_0)(y - y_0)$

$L(x,y,z) = f(x_0,y_0,z_0) + f_x(x_0,y_0,z_0)(x - x_0) + f_y(x_0,y_0,z_0)(y - y_0) + f_z(x_0,y_0,z_0)(z - z_0)$

$\Rightarrow$ in vector form

$L(x,y) = f(x_0,y_0) + \nabla f(x_0,y_0) \cdot (x - x_0, y - y_0)$ → grad f (gradient of f at (x_0,y_0))

$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$

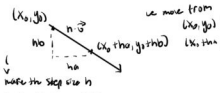
grad $f = (f_x, f_y)$

or in higher dimensions: $\nabla f = (f_x, f_y, f_z)$

Directional derivatives

$$\lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

$$\lim_{\Delta y \to 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}$$



the rate of change is called the directional derivative of f at (x_0, y_0) in the direction of unit vector $\vec{u} = \langle a, b \rangle$

$D_{\vec{u}} f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$

$= \nabla f(x_0, y_0) \cdot \vec{u}$

ex:

$\langle 2, 1 \rangle$ at $(1, \pi)$

$\langle 1, 0 \rangle$ $\langle 0, 1 \rangle$

f_x f_y

$f(x,y) = x \sin y$

$f_x(x,y) = \sin y$ $\Rightarrow f_x(1,\pi) = 0$

$f_y(x,y) = x \cdot \cos y \Rightarrow f_y(1,\pi) = -1$

$\vec{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle 2,1 \rangle}{\sqrt{2^2+1^2}} = \langle 2/\sqrt{5}, 1/\sqrt{5} \rangle$

$D_{\vec{v}} f = \nabla f \cdot \vec{v}$

$= \langle \sin y, x \cos y \rangle \cdot \langle \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \rangle$

$= \frac{2}{\sqrt{5}} \sin y + \frac{1}{\sqrt{5}} x \cos y$ at $(1, \pi)$

$D_{\vec{j}}(1,\pi) = -1/\sqrt{5}$

ex:

$f(x,y,z) = xy\,z^2$

$\langle 2, 1, -2 \rangle$

point $(1, 2, -1)$

$f_x = y\,z^2$ $f_y = x\,z^2$ $f_z = 2xyz$

$\vec{v} = \frac{\langle 2,1,-2 \rangle}{\sqrt{2^2+1^2+2^2}} = \langle 2/3, 1/3, -2/3 \rangle$

$\nabla f = \langle y z^2, x z^2, 2xyz \rangle$

$= \langle (2)(-1)^2, (1)(-1)^2, 2(2)(1)(-1) \rangle$

$= \langle 2, 1, -4 \rangle$

at $(1,2,-1)$

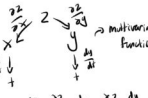
$D_{\vec{v}} f = \frac{2}{3} + \frac{1}{3} + \frac{8}{3} = \frac{11}{3}$

max directional derivative value: $\|\nabla f\|$

Chain rule

$y' = f'(g) \cdot g'(x)$

$\frac{dy}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}$ ⇒ single variable function



⇒ multivariable function

then $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$

ex:

$z = x^2 y^2$

$x = \cos(2t)$ $\frac{dx}{dt} = 2x \cdot -2\sin(2t)$

$y = \sin(2t)$ $+ 2y \cdot 2\cos(2t)$

$= 2 \cdot \cos(2t) \cdot -2\sin(2t) + 2\sin(2t) \cdot \cos(2t) \cdot 2$

$= 0$

$\frac{\partial z}{\partial x} = 2x$

$\frac{dx}{dt} = -2\sin(2t)$

$\frac{\partial z}{\partial y} = 2y$

$\frac{dy}{dt} = 2\cos(2t)$

ex:

at $t = 1$

$z = \cos x^2 + y^3$

$x = \ln t$

$y = t^2 + t + 1$

$\frac{\partial z}{\partial x} = -2x \sin x^2 + y^2$

$\frac{\partial x}{\partial t} = \frac{1}{t}$

$\frac{\partial z}{\partial y} = 2y x$

$\frac{\partial y}{\partial t} = 2t + 1$

$\frac{\partial z}{\partial t} = 4^2 \cdot 1 + 0 \cdot \cdots$

$= 16$

if $z = f(x,y)$

& $x = g(s,t)$, $y = h(s,t)$



$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$

$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$

$\frac{\partial v}{\partial s} = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial v}{\partial z} \cdot \frac{\partial z}{\partial s}$

Implicit differentiation

for single variable function, if we have

$x^2y^2 + 6xy + y^3 = e^x$ & we want $\frac{dy}{dx}$

we did implicit differentiation

$\frac{d}{dx}\left[x^2y^2 + 6xy + y^3\right] = \frac{d}{dx}e^x$

here we don't treat y as a constant because we evaluate $\frac{d}{dx}$ not $\frac{\partial}{\partial x}$

$2xy^2 + x^2 \cdot 2y \frac{dy}{dx} + 6y + 6x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = e^x$

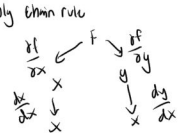
(product rule, product rule, power rule)

$\frac{dy}{dx} = \frac{e^x - 2xy^2 - 6y}{2x^2y + 6x + 3y^2}$

there is another way to do this using chain rule

⇒ if we assume we have a function f of x & y & y is defined implicitly through x $F(x, y(x)) = 0$

if we apply chain rule



$\frac{\partial f}{\partial x}$ $\frac{\partial f}{\partial y}$ $\frac{dx}{dx}$ $\frac{dy}{dx}$

we know, $F(x, y(x)) = 0$, then $\frac{\partial f}{\partial x} = 0$

$\frac{\partial f}{\partial x}\frac{dx}{dx} + \frac{\partial f}{\partial y}\frac{dy}{dx} = 0$

$$\boxed{\frac{dy}{dx} = \frac{-\partial f/\partial x}{\partial f/\partial y} = \frac{-f_x}{f_y}}$$

checks for the previous problem

$f = x^2y^2 + 6xy + y^3 - e^x = 0$

$f_x = \frac{\partial f}{\partial x} = 2xy^2 + 6y - e^x$

$f_y = \frac{\partial f}{\partial y} = 2x^2y + 6x + 3y^2$

⇒ in higher dimensions, if we have f as a function of x, y & z defined implicitly as a function of x & y $F(x, y, z(x,y)) = 0$ then, to get $\frac{\partial z}{\partial x}$, we get $\frac{\partial f}{\partial x} = 0$

$\frac{\partial f}{\partial x}\frac{dx}{dx} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial x} = 0$

$$\boxed{\frac{\partial z}{\partial x} = \frac{-f_x}{f_z}}$$ similarly $$\boxed{\frac{\partial z}{\partial y} = \frac{-f_y}{f_z}}$$

ex: the equation $x^2y + xyz^2 + x = 2z^3$ define z implicitly as a function of x & y. Calculate $\frac{\partial z}{\partial x}$ & $\frac{\partial z}{\partial y}$ at $(1, -1, 0)$

$f = x^2y + xyz^2 + x - 2z^3 = 0$

$\frac{\partial z}{\partial x} = \frac{-f_x}{f_z} = \frac{-2xy - yz^2 - 1}{xy - 6z^2}$ at $(1,-1,0) \Rightarrow \frac{\partial z}{\partial x} = \frac{-2 + 0 - 1}{1 - 0} = -1$

$\frac{\partial z}{\partial y} = \frac{-f_y}{f_z} = \frac{-(x^2 + xz)}{xy - 6z^2}$

at $(1,-1,0) \Rightarrow \frac{\partial z}{\partial y} = \frac{1 + 0}{-1 + 0} = -1$